
Position: Prioritize Identifying Structure, Not Complex Models, for Scientific Discovery

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Abstract

Researchers rarely observe the mechanistic drivers of scientific processes directly. Instead, they observe proxies, which are measurable quantities correlated with one or more of these latent drivers. This position paper argues that mechanistic learning from high-dimensional proxies is generically underdetermined. Many incompatible mechanisms induce essentially the same observational relationships on the support of the data, so predictive success and coherent explanations are insufficient evidence of mechanism discovery. Machine Learning (ML) and Artificial Intelligence (AI) models can therefore generate scientific hypotheses that sound plausible while the underlying mechanism remains unidentified. The problem is particularly acute with large language models (LLMs), which tend to collapse entire equivalence classes of explanations into a single fluent narrative. The paper proposes concrete standards for “Mechanistic AI,” norms that are necessary if AI-centered workflows are to support science rather than merely simulate it.

1. Introduction

Position: Absent explicit *identifying structure*, high-dimensional “scientific discovery” from observed features is generically underdetermined; predictive success and fluent explanations are insufficient evidence of mechanism discovery. Therefore, research should prioritize articulating and evaluating identifying assumptions and discriminating regimes, rather than building ever more complex models. There is a substantial and growing body of work that frames complex AI and ML models as tools for scientific discovery, including symbolic regression (Schmidt

& Lipson, 2009; Udrescu & Tegmark, 2020), sparse identification of dynamics and PDEs (Brunton et al., 2016; Rudy et al., 2017), physics-informed inverse problems (Raissi et al., 2019), and methods that extract interpretable hypotheses from high-dimensional text via sparse autoencoders and LLMs (Movva et al., 2025). In the social sciences, the Fragile Families Challenge tested the value of prediction systems for social outcomes (Salganik et al., 2020), and “agnostic” approaches argue for more inductive, sequential discovery workflows (Grimmer et al., 2021). There is great promise in incorporating AI/ML tools in scientific discovery.

This position paper, however, contends that using modern AI/ML tools agnostically to uncover mechanisms is inherently fraught. Put simply, there are many representations of the relationship between observed features and outcomes that are consistent with any observed dataset. A representation can yield strong in-domain prediction and even convincing simulations, yet it provides no guarantee that the learned representation corresponds to the underlying, intervention-stable driver of the system. The remedy is to make identifying structure explicit. By *identifying structure*, the paper means joint specification of (a) restrictions on the latent mechanism (what the system is allowed to do under interventions) and (b) restrictions on the observation process (how latent drivers generate measured proxies), together with (c) a data-collection regime that supplies discriminating variation (new environments, interventions, or measurement channels). Identifying structure and complex models are not opposites. In many successful scientific-ML systems, complex models are exactly what make it possible to exploit identifying structure once it exists. This point echoes Cartwright’s *nomological machine*: law-like regularities arise when components with stable capacities are arranged in a sufficiently stable environment so that repeated operation yields repeatable behavior (Cartwright, 1999).

The argument starts from a key observation: *scientific workflows rarely observe the drivers of scientific processes directly*. Instead, they observe *proxies* that are associated with one or more latent drivers. In the apocryphal story, Newton didn’t see gravity; he felt a bump on his head from the falling apple. Mendel (1866) observed phenotypes, traits such as seed shape and seed color, recorded for parent plants

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and their offspring. These phenotypes are proxies that, when observed across a set of controlled crosses, gave clues about the underlying scientific mechanism, the transmission of discrete hereditary “factors” across generations. Crucially, Mendel’s success hinged on *identifying structure*, which consists of favorable biological architecture, carefully chosen discrete traits, controlled matings, replication, and a narrow hypothesis class (segregation and independent assortment) against which the proxy patterns could be tested and falsified (Mendel, 1866; Fisher, 1936; Curtis, 2023). The same conclusion could not come from black-box prediction on high-dimensional proxies. For the purpose of this paper, a *mechanistic query* is a concrete question about an underlying data-generating process, such as what would happen under an intervention, whether a relationship is stable across settings, which latent driver explains an observed pattern, or what sign a causal effect has. Such a query $q(M)$ is identified when every mechanism compatible with the observed proxy evidence and stated assumptions gives the same answer, and partially identified when the compatible mechanisms give a bounded or otherwise structured set of possible answers.

The fundamental challenge is that mechanistic inquiry needs to understand both *how good the proxy is* (i.e., how strong the relationship is between the proxy and driving mechanism) and *what it is a proxy for* (i.e., the underlying scientific mechanism). The position here is that, in high-dimensional observational regimes, both cannot be identified without additional structure. The shift from bureaucratic statistical regimes toward brokerage regimes of passively collected digital traces makes this proxy problem increasingly central in social and administrative data (Fourcade et al., 2026). Unstructured discovery, at best, identifies an *equivalence class* of mechanisms based on plausible relationships between the proxies and the mechanisms. Formally, the *proxy observational law* P_X is the population distribution of the observed proxies under the observational regime at hand. For a fixed proxy observational law P_X , let $\mathcal{M}(P_X)$ denote the set of mechanisms and measurement channels that induce P_X . Without identifying structure, $\mathcal{M}(P_X)$ is typically not a singleton. Ogburn et al. (2021) gives examples in the context of causal graphs.

Improving model performance or even the existence of a plausible, consistent explanatory channel is also not sufficient. Coherence is not, by itself, scientific evidence. What’s more, fixating on the wrong mechanism can have dramatic, longstanding consequences. Take, for example, the miasma, or “bad air” theory of disease. The proxies are environmental correlates of illness, such as odor, proximity to swamps/sewers, stagnant water, signs of decaying material, or meteorological conditions. Many of these proxies are strongly correlated with outbreaks, providing seemingly ironclad evidence for the theory. The strength of these as-

sociations, along with the tendency for path dependence in science (Kuhn, 1996; Soler, 2025), meant that the miasma theory remained pervasive even in the presence of mounting evidence for germ-based disease transmission. Mendel, on the other hand, succeeded because favorable biology, selected measurements, and discriminating design effectively made $\mathcal{M}(P_X)$ small for the mechanistic queries he asked. Throughout, the term *design* is used to refer to the structure of the evidence-generating regime, including which variables are measured, which labels or environments are recorded, which naturally occurring contrasts are available, and which interventions or controlled protocols, if any, produced the observations.

A richer feature space might seem like the way out. If you measure enough proxies, then eventually the mechanism should reveal itself. In practice the opposite happens. High-dimensional proxies concentrate on a thin effective support shaped by institutions, selection, and measurement, and as the proxy dimension grows, so does the number of representations consistent with the data on that support. Many incompatible mechanisms can therefore agree closely on the observed data while diverging sharply under interventions or modest domain shifts. Appendix C makes this formal. If P_X is supported on a lower-dimensional set, then off-support behavior is not identified for any rich function class without additional constraints. Xu et al. (2024) prove an “inevitability” result for open-world querying, showing that for any fixed computable learner there exist computable target functions on which it must err on infinitely many inputs. I argue that an analogous inevitability holds for *explanations*. Any body of proxy evidence admits multiple compatible but conflicting mechanistic stories. The identification limits behind this are classical. What LLMs change is the cost, making a fluent, single narrative for any underidentified evidence base one prompt away.

This contention does not depend on a narrow definition of mechanism. The term means stable structural features that explain how a system responds to interventions: derivative constraints (which variables matter and with what sign), conservation laws, invariances, and causal pathways supporting policy counterfactuals (Machamer et al., 2000; Woodward, 2003; Heckman, 2000). Economists distinguish reduced-form associations, which may be accurate yet mechanistically agnostic (Angrist & Pischke, 2010; Athey & Imbens, 2019), from structural models encoding policy-relevant counterfactuals (Heckman, 2000).

The call to action is simple: for ML systems to contribute to scientific discovery, “mechanistic learning” must be treated as an identification problem, with research prioritizing the discovery and declaration of the identifying structure that makes mechanistic questions answerable from proxy data. Accordingly, any mechanistic claim must be

paired with at least one of: (i) a clear statement of this identifying structure (and what remains unidentified), (ii) mechanism-discriminating evaluations that can shrink the proxy-compatible set (interventions, cross-environment invariance tests, derivative/shape constraints), or (iii) multiplicity reporting that characterizes the surviving equivalence class, including explicit falsifiers and sensitivity to the stated assumptions.

The paper proceeds as follows. Section 2 sets up mechanisms, proxies, and models; Section 3 shows that absent identifying structure, proxies identify mechanism equivalence classes rather than unique mechanisms; Section 4 explains how high dimensionality and thin support aggravate those classes; Section 5 introduces LLM narrative collapse; Section 6 relates the ambiguity to aleatoric, epistemic, and structural identification uncertainty; Section 7 gives the empirical example; Section 8 states the call to action; and Section 9 provides alternative views.

2. Problem Setup and Terminology

Let S denote latent scientific drivers (“mechanism-level” variables). The observed data consist of proxies X generated from S by a measurement process. Consider deterministic measurement $X = f(S)$ and stochastic measurement channels $X | S \sim K(\cdot | S)$, where K is a Markov kernel from the latent space to the proxy space, covering additive noise, discretization, aggregation, censoring, and other coarsenings (Fuller, 1987; Carroll et al., 2006). Given a latent law P_S and measurement channel K , the induced *proxy observational law* is $P_X(A) = \int K(A | s) P_S(ds)$ for measurable proxy events A . The *proxy gap* is the distinction between this observable law and the latent mechanism $M = (P_S, K)$, or a mechanistic query $q(M)$, when distinct mechanisms induce the same P_X . A modern workflow trains a predictor h , often through a learned representation $Z = \phi(X)$, to optimize an in-domain objective, such as predicting an outcome Y from proxies X , which is distinct from recovering the latent S .

To move from proxies to mechanisms, many methods pool “similar” observations, either explicitly (clustering, trees, mixtures, nearest neighbors) or implicitly (representation learning that induces neighborhood smoothing). The workhorse assumption behind pooling is a symmetry choice often framed as *exchangeability*: declaring certain differences irrelevant so that averaging is meaningful. Exchangeability yields powerful representation theorems (de Finetti, 1964; Aldous, 1981; Kallenberg, 2005; Diaconis & Freedman, 1980). But representation is not identification: the theorems specify what must be true *if* a symmetry holds, but do not guarantee the symmetry is correct, nor that a unique mechanism is pinned down by proxies. Kleinberg’s clustering impossibility theorem makes this concrete (Klein-

berg, 2002). This paper makes three claims. Proofs of the three forthcoming Propositions are in Appendix B. For clarity, the objects relate as:

$$S \xrightarrow[\text{measurement}]{K \text{ or } f} X \xrightarrow[\text{encoding}]{\phi} Z \xrightarrow[\text{predictor}]{h} \hat{Y},$$

Many predictors or encodings can map the same proxies to similarly good predictions, but that happens to the right of X . The proxy gap lives to the left of X : many latent mechanisms and measurement channels can generate the same proxies while implying different mechanistic answers. Identification concerns what can be inferred about S or $q(M)$ from X , not merely how well h predicts Y in-domain.

3. Claim I: The Proxy Gap Creates Mechanism Equivalence Classes

Proxies constrain mechanisms only through a many-to-one measurement process (deterministic or stochastic). Without additional identifying structure, the evidence generally identifies an *equivalence class* of mechanisms rather than a unique mechanism. Causal-quartet examples in D’Agostino McGowan et al. (2024) show this in the context of causal inference. For 19th-century epidemic disease, the day-to-day evidence available to observers—symptoms, odors, local air conditions, seasonality, crowding, sanitation, neighborhood mortality rates, and water/waste correlates—was often compatible with both miasmatic and germ-theoretic accounts. Miasmatic theories were empirically plausible because foul environments were genuinely predictive of disease, and sanitation often improved health; germ-theoretic accounts became compelling only when new evidence decoupled these proxies from specific microbial exposure and transmission mechanisms. With only the original proxy channel and no new measurement or identification restriction, observing more cases need not resolve the mechanistic ambiguity.

The argument begins with an intentionally “easy” setting. Assume the latent state decomposes into B blocks $S = (S_1, \dots, S_B)$. A *proxy block* X_b is the collection of observed measurements intended to capture (possibly noisily) the corresponding latent block S_b . Assume a modular measurement structure in which each proxy block depends only on its corresponding latent block (e.g., multiple sensors targeting the same driver), either deterministically $X_b = f_b(S_b)$ or via a channel $X_b | S_b \sim K_b(\cdot | S_b)$. The propositions below show that even if this modular structure holds and measurement is invertible within blocks, proxies alone still do not uniquely identify the latent mechanism without additional identifying assumptions or discriminating regimes.

Proposition 1 (Proxy identifiability) *Let $X = (X_1, \dots, X_B)$ be observed proxy blocks. Suppose*

an admissible deterministic representation consists of latent blocks $S = (S_1, \dots, S_B)$ and bimeasurable bijections $f_b : S_b \rightarrow \mathcal{X}_b^{\text{supp}}$ such that $X_b = f_b(S_b)$ a.s. for each $b = 1, \dots, B$. If a second admissible representation of the same proxy blocks is given by $X_b = \tilde{f}_{\pi(b)}(\tilde{S}_{\pi(b)})$ a.s. for each b , where π is a permutation of $\{1, \dots, B\}$ and each $\tilde{f}_{\pi(b)}$ is also a bimeasurable bijection onto $\mathcal{X}_b^{\text{supp}}$, then there exist bimeasurable bijections satisfying, for each b ,

$$g_b := \tilde{f}_{\pi(b)}^{-1} \circ f_b, \quad \tilde{S}_{\pi(b)} = g_b(S_b) \text{ a.s.}, \\ \tilde{f}_{\pi(b)} = f_b \circ g_b^{-1}.$$

In this deterministic invertible setting, the equivalence class induced by P_X is exactly within-block reparameterizations plus, when labels are not intrinsic, block permutations. Thus only quantities invariant under those transformations (e.g., certain conditional independences or invariance relations) are identified from P_X without further structure. This population-level ambiguity is not a finite-sample issue and does not disappear as parameter uncertainty shrinks; Proposition 2 shows it also survives arbitrary conditionally independent measurement noise.

Proposition 2 (Non-identification persists under noise)

Let $S = (S_1, \dots, S_B)$ and let $K(x | s) = \prod_{b=1}^B K_b(x_b | s_b)$ be a conditionally independent proxy channel. For any blockwise bimeasurable maps g_b , define

$$\tilde{S}_b = g_b(S_b), \quad \tilde{K}_b(\cdot | \tilde{s}_b) := K_b(\cdot | g_b^{-1}(\tilde{s}_b)).$$

Then (S, K) and $(\tilde{S}, \tilde{K})$ induce the same distribution of X . Thus, blockwise reparameterization non-identification survives arbitrary conditionally independent measurement noise.

Richer proxies sharpen *estimation* within a chosen model. *Identification* of a mechanism is a different problem, and richer proxies alone do not solve it.

4. Claim II: High Dimensionality Aggravates the Equivalence-Class Problem

The proxy gap already creates mechanism equivalence classes. High dimensionality makes those classes bigger and harder to see. High-dimensional proxy vectors rarely fill their ambient space. They concentrate near a much lower-dimensional set determined by how the data were selected, measured, and correlated, and the curse of dimensionality means that learning general functions (let alone mechanism-relevant counterfactual behavior) requires sample sizes that explode with dimension unless strong structure is assumed (Bellman, 1957; Stone, 1982).

Geometry drives this phenomenon. In high dimensions distances concentrate, so “nearest” and “farthest” neighbors

can become nearly indistinguishable under broad conditions (Beyer et al., 1999; Aggarwal et al., 2001). Once neighborhoods lose meaning, any method that pools locally (nearest neighbors, kernels, clustering, representation-induced smoothing) becomes sensitive to modeling choices that the data cannot adjudicate. Partitioning compounds the problem. Even a coarse partition of a d -dimensional proxy space has exponentially many cells, and unless sample sizes grow to match, most cells contain too little data to distinguish mechanisms that agree where the data sit but differ everywhere else. This is what a large equivalence class looks like in finite samples: rival mechanisms fit the observed support equally well, and the regions that would separate them are empty.

In sum, Sections 3–4 argue that in proxy-rich, high-dimensional regimes, mechanistic claims should be treated as claims about a *set* of observationally compatible mechanisms unless accompanied by identifying structure or mechanism-discriminating evaluation.

5. Claim III: An Additional LLM-Specific Hazard—Narrative Collapse as False Resolution

Sections 3–4 argue that, without identifying structure, proxy evidence supports a set of observationally compatible mechanisms rather than a unique mechanism. The trouble starts when an LLM-centered workflow presents that set as settled. I call this *narrative collapse*, a many-to-one mapping from a mechanism compatibility set to an explanatory output that reads as more identified than the evidence warrants. The output might be a single story, a ranked list of factors, or a polished synthesis. Scientific communities have always converged on “the” story under ambiguity. What is different now is that an LLM produces the single story in seconds, on demand, for any dataset put in front of it. What’s more, the story is compelling by design.

Many systems, especially when prompted, list several factors and caveats. A factor list still collapses the science unless it preserves the multiplicity that matters: which mechanisms are mutually incompatible, which assumptions make each admissible, which answers to $q(M)$ are stable across $\mathcal{M}(P_X)$, and what evidence would distinguish them.

This use of “collapse” is intentionally different from two nearby concerns. *Model collapse* refers to degenerative dynamics that can arise when generative models are trained recursively on model-generated content, leading to distributional narrowing and loss of tails (Shumailov et al., 2024). *Algorithmic monoculture* refers to population-level harms when many decision-makers converge on the same algorithm, reducing system robustness even when the algorithm is locally better for each agent (Kleinberg & Raghavan,

2021). Both phenomena can amplify narrative collapse, but neither is required for it: narrative collapse can occur in a single interaction with a single model on a fixed dataset, whenever the evidence does not identify a unique mechanism and the interface reports a resolved explanation without representing what remains compatible.

How often current systems collapse in this way is an empirical question, and the answer will vary with model, prompt, interface, and task. The argument is conditional. Whenever a workflow asks for, or rewards, a resolved mechanistic answer while $\mathcal{M}(P_X)$ contains mechanisms with different values of $q(M)$, no such answer can be uniformly correct. Interfaces that surface the compatible set, tag assumptions, and propose discriminating tests reduce the risk. Building those interfaces is part of the call to action in Section 8.

To make the issue precise without over-formalizing, fix an observed proxy distribution P_X and let $D = (X^{(1)}, \dots, X^{(n)}) \sim P_X^n$ denote the observed dataset. Let $\mathcal{M}(P_X)$ denote the set of mechanisms (latent data-generating processes for S together with measurement processes K) that are observationally compatible with P_X under the setup of Sections 2–3. Let $q : \mathcal{M} \rightarrow \mathcal{A}$ be a mechanistic query mapping a mechanism to an answer space $\mathcal{A}$ (e.g., “Which variables are causally upstream?” “What intervention increases Y ?” “Which invariance should hold out of domain?”). Everything below holds verbatim if D includes outcomes Y or other observables: simply replace P_X by the joint observational law of whatever variables the workflow conditions on. Proposition 3 records the basic obstruction: if two proxy-compatible mechanisms imply different answers to q , then no rule that sees only the proxies can always give the right single answer.

Proposition 3 (Narrative collapse as minimax ambiguity)

If there exist $M_1, M_2 \in \mathcal{M}(P_X)$ such that $q(M_1) \neq q(M_2)$, then no single-valued explanation rule $\hat{a} = \hat{a}(D)$ can be uniformly correct over $\mathcal{M}(P_X)$ for the query q . In particular, under 0–1 loss $\ell(\hat{a}, q(M)) = \mathbf{1}\{\hat{a} \neq q(M)\}$, the worst-case risk satisfies

$$\inf_{\hat{a}} \sup_{M \in \mathcal{M}(P_X)} \mathbb{E}[\ell(\hat{a}(D), q(M))] \geq \frac{1}{2},$$

where the expectation is over $D \sim P_X^n$ (equivalently, over any procedure that only has access to the proxies).

Proposition 3 is deliberately illustrative: when the evidence admits multiple incompatible answers to a mechanistic query, any *single* reported answer must be wrong for at least one observationally compatible mechanism. The scientifically responsible response is therefore to change what is reported, rather than how it is reported. In interactive use, the risk is amplified when follow-up questions move from description to “why” and “what-if” claims. Those questions

can require extrapolating beyond what the proxies identify. Conversational norms can also reward *consistency*. Once a workflow commits to a mechanism in an early turn, later turns may rationalize and elaborate that commitment rather than reopening the mechanism set. This coherence pressure is useful for ordinary assistance, but under underidentification it can turn a set-valued scientific state of knowledge into an increasingly entrenched point narrative.

Narrative collapse converts a set-valued state of scientific knowledge into a single, resolved explanatory object. It is the mechanism-level analogue of overconfident extrapolation. Evaluating it requires benchmarks that pair proxy data with a mechanistic query q , the set of observationally compatible answers under stated assumptions, and at least one discriminating test that would shrink that set. Against such a benchmark, how many causes a response names matters less than whether it covers the compatible answers, keeps mutually incompatible mechanisms separate, tags its assumptions, and makes point claims only once the evidence identifies the query.

6. Where This Sits Relative to Aleatoric and Epistemic Uncertainty

Uncertainty quantification in ML is often organized around a dichotomy: *aleatoric* uncertainty, attributed to irreducible randomness in outcomes, and *epistemic* uncertainty, attributed to an agent’s lack of knowledge about parameters, models, or hypotheses (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). This split is useful for prediction problems, but it does not cleanly capture the central phenomenon in Sections 3–5. Even with unlimited data about proxies X , mechanism-level claims about S can remain underdetermined unless one adds explicit identifying structure. This paper isolates a particularly important form of epistemic uncertainty: uncertainty induced by non-identification from proxies.

To make this precise (with a more formal presentation in Appendix D), fix an observational distribution P_X . Consider the set of mechanism stories that could have produced it: $\mathcal{M}(P_X) = \{(P_S, \text{measurement map/channel}) : (S \rightarrow X) \text{ induces } P_X\}$. Sections 3–4 argue that $\mathcal{M}(P_X)$ is typically large in modern regimes: proxies are many-to-one summaries of latent drivers, and high dimensionality makes disagreement outside the observed support easier to hide and harder to diagnose. This is the *structural identification uncertainty* emphasized here. In the context of causal inference, Gelman et al. (2024) show that even when a scalar average causal effect is held fixed, very different heterogeneous effect patterns can remain compatible with that same summary, for example. In contrast, aleatoric uncertainty concerns variability of outcomes *conditional on a mechanism*, while common uses of epistemic uncertainty quantify

uncertainty relative to a chosen model or hypothesis class. The proxy gap matters because it questions whether that class and its measurement geometry identify the mechanistic query at all.

This perspective clarifies how these claims relate to (and differ from) the Rashomon effect and underspecification. Rashomon-style multiplicity concerns many predictors achieving similar risk (Fisher et al., 2019; Xin et al., 2022; Venkateswaran et al., 2024); underspecification emphasizes that many models can fit training objectives yet behave differently under deployment (D’Amour et al., 2022). Structural identification uncertainty is more fundamental. From X alone, the mapping back to the mechanism S is not pinned down without additional assumptions about measurement and structure. Rashomon and underspecification are important amplifiers in finite samples and deployment, but the proxy gap can generate mechanism equivalence classes even before optimization and architecture enter the picture.

These distinctions matter for LLM-centered scientific workflows. Many uncertainty quantification (UQ) tools quantify dispersion around a well-defined predictive target. But when the user asks a mechanism question—“what explains this pattern?”—the dominant uncertainty may be which members of $\mathcal{M}(P_X)$ remain plausible. In that regime, a single calibrated interval or scalar “epistemic uncertainty” can be actively misleading: it suggests that what is missing is merely more data or better parameter estimates, when the constraint is lack of identification structure. This issue is sharper in interactive settings, where queries expand into counterfactuals and the system is forced to pick a story off-support; recent critiques argue that the aleatoric/epistemic split becomes strained for LLM agents in open interaction (Kirchhof et al., 2025). Even perfect predictive calibration can leave mechanistic claims non-identified.

7. An Example with Pea Plants

Mendel crossed a pure-breeding wrinkled-green line with a pure-breeding round-yellow line. The first generation (F_1) was uniformly round-yellow, and those F_1 plants were crossed again to produce the F_2 generation analyzed here. The observed labels are visible traits—round-yellow (RY), round-green (RG), wrinkled-yellow (WY), and wrinkled-green (WG)—not the hereditary state itself. The hidden mechanism is the rule by which parental hereditary factors produce those visible categories. Mendel worked with unusually discrete, selected, experimentally tractable traits; many genetic traits involve many loci, interactions, pleiotropy, and environmental dependence (Curtis, 2023; Bapty, 2023; Mackay & Anholt, 2024). This section asks what happens when that structure is removed from the evidence regime.

The simulation is designed to make three points concrete. First, under a phenotype-only observational channel, distinct mechanisms can be observationally equivalent on the support of the data (Claim I). Second, adding experimental structure (here, labeling the cross type and including additional crosses) shrinks this mechanism equivalence class (Claim II). Third, modern “black-box discoverers” can achieve similar predictive performance while still yielding unstable mechanistic counterfactuals when the design metadata is removed (Claim III). The simulation compares two regimes. Regime A gives only pooled F_2 phenotype counts (the offspring categories). Regime B also records which controlled cross produced each observation (`F2_self`, `testcross`, `monoA`, or `monoB`) and includes additional labeled crosses. This cross label is the identifying structure since phenotype counts alone record what the offspring looked like, but the mating protocol tells us whether those counts came from an F_2 self-cross, a testcross, or a single-locus cross, which are different tests of the inheritance rule.

The example compares two stories that look identical if only the pooled F_2 counts are inspected. The Mendelian story says those counts come from a real inheritance rule. The rule combines two traits, dominance, and independent assortment, which together produce the familiar 9:3:3:1 pattern. The deliberately simple alternative ignores that inheritance rule and instead treats shape and color as two separate weighted coin flips, each producing the dominant trait with probability $3/4$. This alternative is constructed to match the pooled F_2 counts exactly. But once the data record which cross produced each offspring, the two stories no longer agree. In *Regime A* (phenotype-only), the simulation generates $n_{F_2} = 6000$ offspring from an F_2 -self distribution (parents $RY \times RY$), and the analysis evaluates fit only at the level of the pooled phenotype distribution. In *Regime B* (design), the simulation generates additional labeled crosses, each with $n_{\text{per cross}} = 1200$ offspring: `F2_self`, `testcross`, `monoA`, and `monoB`. The key identifying structure is the cross-type label (e.g., `testcross` vs `mono` cross), which encodes experimental lineage/design information that is not recoverable from phenotype proxies alone but radically changes what the same proxy observations imply. The analysis fits a two-hidden-layer Multilayer Perceptron (MLP) classifier with architecture (32, 32) to predict offspring phenotype from parent phenotypes. In the figure labels, “no design” is Regime A: the MLP trains without cross labels and sees only F_2 self-cross data. “With design” is Regime B: the MLP trains with the cross label and sees the queried testcross support. A seed is one independently trained MLP run with a different random-state value controlling random initialization and training randomness; the train/test split is fixed across seeds. The “all” bars use all 60 such runs in the relevant regime; the “near-opt” bars restrict to seeds whose

held-out test log loss is within $\varepsilon = 0.01$ of the best seed for that same regime.

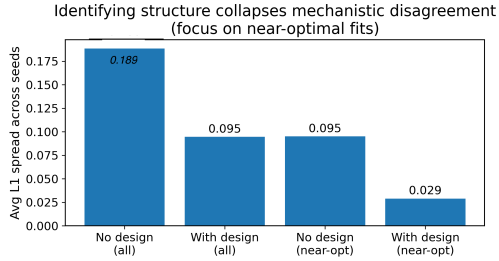


Figure 1. Identifying structure collapses mechanistic disagreement. Mean ℓ_1 spread of counterfactual predicted probability vectors across MLP runs. “No design” is Regime A (phenotype-only F_2 self-cross data); “with design” is Regime B (cross labels plus additional labeled crosses, including testcross support). “All” uses all 60 random-state runs; “near-opt” restricts to runs within $\varepsilon = 0.01$ of the best held-out test log loss in the same regime. The spread collapses in the design regime, which adds testcross support and labels that make the counterfactual cross prediction well defined.

The prediction task asks, for the same observed parent phenotype pair $RY \times WG$, what offspring probabilities the model should assign to RG , RY , WG , and WY . Without cross labels, this question is underdetermined because the same parent phenotypes can arise from different experimental lineages; with labels, it becomes a specific testcross prediction (`cross=testcross`). The mean ℓ_1 spread summarizes how much the fitted models disagree: $\frac{1}{N_{\text{seeds}}} \sum_{r=1}^{N_{\text{seeds}}} \|\hat{p}_r - \bar{p}\|_1$, where $\bar{p} := \frac{1}{N_{\text{seeds}}} \sum_{r=1}^{N_{\text{seeds}}} \hat{p}_r$, computed either across all seeds or only near-optimal seeds. Zero would mean every fitted network gives the same four probabilities. Figure 1 shows the point. In Regime A, equally good predictors disagree more about the counterfactual cross; in Regime B, the added cross labels and testcross support make the answer substantially more stable. The near-optimal spread falls from 0.095 without design to 0.029 with design, while the all-seed spread falls from 0.189 to 0.095. Additional concrete probabilities and plots are in Appendix E; a continuous low-dimensional simulation is in Appendix F.

8. Call to Action: Mechanistic AI

The call is to treat *identification*, rather than expressiveness, as the binding constraint in proxy-rich scientific discovery. By identification I mean pinning down the answer to a stated mechanistic query within a compatibility set of mechanisms, which does not require a randomized or quasi-random treatment assignment. First-principles restrictions are common examples. Conservation laws, symmetries, monotonicity, geometry, and intervention semantics all shrink the compatible mechanism set and reduce the ability of alternative representations to produce similar observed patterns. Identifi-

cation is a common concept in economics and causal inference, among other areas. For example, unsupervised disentanglement is provably non-identifiable without inductive biases and/or supervision (Locatello et al., 2019), while identifiability can be recovered under additional observed variables, grouping structure, or restricted ICA-style assumptions (Khemakhem et al., 2020; Morioka & Hyvarinen, 2024). More broadly, the logic of no-free-lunch and Kolmogorov complexity arguments makes the same meta-point. Without inductive bias aligned with the world, learning cannot select the right explanation class (Goldblum et al., 2024). In the presence of proxies, Mechanistic AI should adopt the same norms in pursuit of scientific discovery. The central question is “under what assumptions and what data design is the mechanistic query $q(M)$ identified?” This framing contrasts with an emphasis solely on flexibility in replicating patterns that appear in the data. Appendix G gives six examples.

Appendix H defines identifying structure with mathematical formalism, while Appendix I gives an example of how this conceptualization can be used to identify minimal identification assumptions for a specific class of models. This is also why several successful scientific-ML systems should be viewed as positive cases rather than counterexamples. Equation-discovery and inverse-problem methods such as symbolic regression, SINDy, and PINNs succeed when they combine expressive function classes with explicit structural restrictions on the mechanism, the observation process, or the experimental regime. Their success comes not from black-box flexibility alone, but from the identifying structure. In practice, there are several indicators of such structure. First, a workflow can *declare assumptions* that restrict both the mechanism and the measurement channel, so that readers can see which equivalence classes have been ruled in and ruled out. Second, it can *collect and evaluate on discriminating data* by performing interventions, observing invariances across environments, and deriving constraints implied by the mechanism but not by proxy correlations alone. This is where design enters with a call for *new* data that breaks observational equivalence by changing environments, manipulating inputs, or measuring closer to the latent drivers. Without a notion of identifying structure, there is no guidance about *how* to collect data. Third, when assumptions and available data still leave ambiguity, a workflow can *report multiplicity* by characterizing the remaining identified set of mechanistic answers (or mechanisms) rather than collapsing it into a single narrative.

Mendel’s case worked because favorable biology and discriminating design lined up. His hypothesis class was sharply constrained (particulate inheritance with segregation and independent assortment), his selected discrete phenotypes tracked the latent hereditary factors stably enough to yield falsifiable implications, and his controlled crosses,

replication, and targeted contrasts such as F_2 ratios and testcrosses would have come out differently under rival mechanisms (Mendel, 1866; Curtis, 2023). Design did not make genetics simple. It made rival mechanisms imply different observable patterns, which is what mechanistic discovery requires of an evidence regime. Miasmatic theories of epidemic disease failed exactly this test. Their proxies were rich but indirect, they predicted outbreaks well, and sanitary reforms motivated by miasmatic reasoning often improved health, so the theory stayed empirically plausible for decades. The same proxy patterns were equally compatible with germ-theoretic accounts, because waste, water, housing, and occupational environments also structured microbial exposure and host susceptibility. Sorting the two apart took evidence that decoupled the mechanisms. Water-source and exposure contrasts in cholera, laboratory isolation and culture, experimental inoculation, and later microbiological assays (Snow, 1855; Pasteur, 1861; Koch, 1884). The demand on Mechanistic AI is the same. Choose data to shrink $\mathcal{M}(P_X)$. More data from the same channel cannot do it (Corollary 1).

The following expectations should apply to any workflow, especially an LLM-centered one, that presents outputs as mechanistic. First, **Mechanism cards**: authors should combine every mechanistic explanation with a short structured appendix that (i) states the *mechanism-side* assumptions (what is invariant under interventions, which functional forms/constraints are being imposed, what is excluded), (ii) states the *observation-side* assumptions (how proxies relate to latent drivers; what is assumed stable vs. artifactual in measurement), and (iii) states what the assumptions *actually identify*, namely which aspects of the mechanism are pinned down and which remain an equivalence class (Appendix H). A mechanism card is incomplete without at least one explicit falsifier consisting of an additional environment, intervention, new measurement channel, or invariance/derivative test that would *rule out* the proposed mechanism. Second, **Multiplicity statements** require that when identification is unavailable, the system must represent the identified set rather than output a single story. Concretely, it should report a set of distinct compatible mechanisms or a set of answers to the query q (e.g., a sign-robust derivative range, a range of counterfactual effects), and it should label which qualitative conclusions are stable across that set. Third, **Mechanism-discriminating evaluation**. Mechanistic claims must be evaluated on targets that directly probe identifying structure like interventions, invariances across environments, proxy-invariance checks tied to measurement, and derivative constraints.

If a paper or product claims a black box has “discovered a mechanism,” reviewers and readers should ask what is assumed about the mechanism, what is assumed about the measurement channel, what data or contrasts actually dis-

criminate among alternatives, and what else remains compatible. If the answer is unclear, the authors should phrase the results as a coherent hypothesis, simulation, or partially identified set consistent with the observed proxies on their realized support, together with a clear statement of what the evidence rules out and what remains compatible. In these settings, human experts remain central because the key tasks are to state defensible assumptions, design discriminating experiments, and decide which claims are prediction-only versus mechanistic.

9. Alternative Views

Alternative View 1: “This is just epistemic uncertainty; ensembles and calibration solve it.” Ensembles and calibration are valuable, but they primarily address uncertainty about predictions under a fixed observational target distribution. They do not, by themselves, identify mechanisms when proxies admit multiple observationally compatible causal stories (Sections 3–4). The relevant ambiguity here is *structural identification uncertainty*: even with unlimited proxy data from the same measurement channel, mechanistic queries $q(M)$ can differ across $M \in \mathcal{M}(P_X)$. Diversity helps when it *surfaces* disagreement, but that requires an interface that does not collapse into one story.

Alternative View 2: “Scaling and better data will select the true mechanism.” This objection is partly right. Additional data help *when they add mechanistically informative variation*—new environments, interventions, or measurement channels closer to the latent drivers—because these can shrink $\mathcal{M}(P_X)$. However, scale alone can also improve in-domain fit while leaving $\mathcal{M}(P_X)$ large. In this framework, this objection is best understood as a claim about the observational regime r : if scaling implicitly changes r (e.g., by adding environments or new measurement channels), then it is adding identifying structure. The request is that such structure be made explicit and evaluated via discriminating tests, rather than assumed. Corollary 1 makes this precise: if two observationally compatible mechanisms disagree on q , no increase in sample size within the same observational regime can point-identify q .

Alternative View 3: “Identification is too strong a standard; useful science often proceeds with partial evidence.” This objection is right, and it is part of the position here. Many scientific and policy tasks do not require point-identifying a general mechanism. Partial identification, triangulation, process tracing, negative controls, and falsification tests can rule out candidate explanations, bound plausible effects, shift beliefs, guide data collection, and justify provisional action under uncertainty. The standard advocated here is narrower. When a workflow claims to have discovered a generalizable, intervention-stable mecha-

nism, it should either supply the identifying structure that makes that claim answerable or report the remaining multiplicity. Evidence short of identification is valuable when it is labeled as such.

Alternative View 4: “Narrative collapse is a UI issue, not a scientific issue.” Narrative collapse is partly an interface issue. When the interface encourages single-story explanations in underidentified settings, it invites overclaiming. Related phenomena like model collapse and algorithmic monoculture can amplify the problem in ecosystems, but narrative collapse can occur even in a single interaction on a fixed dataset. The appropriate response is reporting and evaluation norms that force mechanistic claims to be conditional on discriminating tests and to represent multiplicity when identification is unavailable.

Alternative View 5: “Occam/simplicity will pick the right mechanism among observationally equivalent ones.” Simplicity biases (explicit or implicit) can be effective inductive biases in practice, and they often guide scientific progress. The point here is that simplicity is itself an *identifying assumption*: it selects one member of a proxy-compatible equivalence class. When a workflow reports a single mechanism on the basis of simplicity, it should declare (i) the simplicity criterion being used (description length, sparsity, functional form, etc.), (ii) which alternatives remain compatible absent that criterion, and (iii) falsifiers or regime shifts that would discriminate between the selected mechanism and plausible competitors.

Alternative View 6: “Mechanisms are only defined up to query-specific equivalence; reporting one representative is fine.” Mechanism claims are indeed query-specific: if two mechanistic stories imply the same answer to the intended interventional query q , then they are equivalent *for that query*. The concern is that LLM-centered workflows tend to output a rich narrative that contains many *additional* mechanistic commitments not warranted by the evidence (or by the intended query), and to present those commitments as uniquely supported. A safer norm is to separate (i) query-stable conclusions from (ii) narrative embellishments that vary across $M \in \mathcal{M}(P_X)$.

When ML/LLM-centered workflows *can* genuinely support discovery. This paper argues against treating fluent narratives or predictive success as evidence of mechanism discovery absent identifying structure, not against complex models. AlphaFold2, for example, pairs model complexity with a narrow target query, exact sequence inputs, evolutionary signal from alignments/templates, experimentally determined training structures, geometry-aware inductive biases, and blind evaluation against withheld structures (Jumper et al., 2021; The Nobel Prize, 2024). LLM-centered work-

flows can be constructive when they generate candidate mechanisms with explicit identifying assumptions, propose discriminating experiments, measurements, or invariance tests that shrink $\mathcal{M}(P_X)$, and summarize evidence as identified sets rather than single narratives when identification is unavailable. Relatedly, recent work on open-ended autonomous discovery proposes steering hypothesis search by Bayesian surprise (Agarwal et al., 2025); the concern here is orthogonal. Even principled exploration must either represent multiplicity or introduce discriminating structure when the same observational channel admits multiple incompatible mechanisms.

10. Conclusion

Without explicit identifying structure, it is not possible to tell whether a fluent model has learned a mechanism or merely a representation. The position here would be weakened by a demonstrated class of proxy-rich observational regimes in which (i) mechanistic claims can be reliably distinguished, tightly bounded, or otherwise made robust using only declared observational structure, and (ii) LLM-centered workflows reliably surface those limits, bounds, and falsifiers rather than collapsing to a single narrative. High-performing predictive models can still be scientifically valuable for emulation, screening, design, measurement, and hypothesis generation even when they are not mechanistically faithful. Predictive or descriptive success should not be automatically elevated into mechanism discovery, though. Absent such evidence, we should build models that predict well, and require mechanistic claims to be stated in a form that supports falsification, sensitivity analysis, and explicit accounting of multiplicity. Otherwise, the field may well end up “discovering” the next miasma.

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A. Summary Table of Running Examples

Table 1 gives a summary table of the two examples used repeatedly throughout the paper.

Case	Observed proxies X	Latent factors / mechanism S	Identifying structure (what shrinks $\mathcal{M}(P_X)$)
Mendel (peas)	Discrete phenotypes (e.g., seed color/shape, flower color) across controlled crosses	Genotypes/alleles; segregation and (approx.) independent assortment; genotype→phenotype mapping	Designed experiments (controlled mating, factorial crosses), targeted contrasts (F_2 ratios, test-crosses), and low-dimensional mechanistic parameterization that makes alternatives falsifiable
Miasma vs germ-theoretic causation	Environmental correlates: odor/“bad air”, visible fog, proximity to swamps/sewers, neighborhood incidence, weather/wind; coarse spatial-temporal aggregates	Specific pathogen presence and transmission route (e.g., water-borne contamination), contact networks/exposure pathways; heterogeneous susceptibility	Mechanism-discriminating measurements and contrasts: water-source comparisons, exposure histories, quasi-experimental exposure changes, microbiological assays, and environment shifts that decouple proxies from causes

Table 1. Running examples: proxies, latent mechanisms, and the extra structure required to identify mechanisms rather than merely fit observations.

B. Proofs of Propositions 1–3

Proof of Proposition 1 (Proxy identifiability up to blockwise reparameterization). Fix two admissible deterministic representations of the same proxy blocks:

$$X_b = f_b(S_b) \quad \text{a.s.} \quad \text{and} \quad X_b = \tilde{f}_{\pi(b)}(\tilde{S}_{\pi(b)}) \quad \text{a.s.}$$

for each $b = 1, \dots, B$, where π is a permutation of $\{1, \dots, B\}$ and each f_b and $\tilde{f}_{\pi(b)}$ is a bimeasurable bijection onto $\mathcal{X}_b^{\text{supp}}$.

For each block b , define

$$g_b := \tilde{f}_{\pi(b)}^{-1} \circ f_b.$$

Because f_b and $\tilde{f}_{\pi(b)}$ are bimeasurable bijections, g_b is a bimeasurable bijection from $\mathcal{S}_b$ onto $\tilde{\mathcal{S}}_{\pi(b)}$, with inverse

$$g_b^{-1} = f_b^{-1} \circ \tilde{f}_{\pi(b)}.$$

Applying $\tilde{f}_{\pi(b)}^{-1}$ to the two representations of the same observed proxy block X_b gives

$$\tilde{S}_{\pi(b)} = \tilde{f}_{\pi(b)}^{-1}(X_b) = \tilde{f}_{\pi(b)}^{-1}(f_b(S_b)) = g_b(S_b) \quad \text{a.s.}$$

for each b . Rearranging the definition of g_b yields

$$\tilde{f}_{\pi(b)} = f_b \circ g_b^{-1}.$$

Therefore any two admissible deterministic representations of the same proxy blocks differ only by blockwise bimeasurable reparameterizations, together with a permutation when the block labels are unlabeled.

Finally, because every admissible deterministic representation of an observational law P_X can be realized by taking $X \sim P_X$ and setting $S_b = f_b^{-1}(X_b)$ blockwise, the same conclusion characterizes the deterministic observational equivalence class associated with P_X . Hence, within this admissible deterministic class, the latent blocks are identified exactly up to within-block bimeasurable reparameterization and block permutation. $\square$

Why is this a problem? Without additional anchors, the coordinates of a recovered latent block have no stable scientific meaning by themselves: S_b and $\log S_b$ can represent the same evidence under a compensating measurement map, yet

imply different claims about linearity, additivity, or intervention magnitude. Only reparameterization-invariant features are identified absent further structure.

Proof of Proposition 2 (Non-identification persists under noise). Let P_S denote the joint distribution of $S = (S_1, \dots, S_B)$. The proxy distribution induced by (P_S, K) is, for any measurable A in the proxy space,

$$P_X(A) = \int K(A | s) P_S(ds).$$

Assume $K(x | s) = \prod_{b=1}^B K_b(x_b | s_b)$ (conditionally independent channels). Let $g(s) := (g_1(s_1), \dots, g_B(s_B))$ with each g_b bimeasurable, and define the reparameterized latent variable $\tilde{S} := g(S)$ (so $P_{\tilde{S}} = P_S \circ g^{-1}$). Define the pulled-back channels

$$\tilde{K}_b(\cdot | \tilde{s}_b) := K_b(\cdot | g_b^{-1}(\tilde{s}_b)), \quad \text{and} \quad \tilde{K}(\cdot | \tilde{s}) := \prod_{m=1}^B \tilde{K}_b(\cdot | \tilde{s}_b).$$

Because each g_b^{-1} is measurable and each K_b is a Markov kernel, each $\tilde{K}_b$ is again a Markov kernel. Then for any measurable set A ,

$$\tilde{P}_X(A) := \int \tilde{K}(A | \tilde{s}) P_{\tilde{S}}(d\tilde{s}).$$

Using the defining property of a pushforward measure,

$$\int \varphi(\tilde{s}) P_{\tilde{S}}(d\tilde{s}) = \int \varphi(g(s)) P_S(ds) \quad \text{for any measurable } \varphi,$$

and taking $\varphi(\tilde{s}) = \tilde{K}(A | \tilde{s})$ yields

$$\tilde{P}_X(A) = \int \tilde{K}(A | g(s)) P_S(ds) = \int K(A | s) P_S(ds) = P_X(A),$$

where the middle equality holds because

$$\tilde{K}(A | g(s)) = \prod_{b=1}^B \tilde{K}_b(A_b | g_b(s_b)) = \prod_{b=1}^B K_b(A_b | s_b) = K(A | s)$$

(first for measurable rectangles $A = \prod_b A_b$, then extended to the full σ -algebra by standard monotone class arguments). Hence the induced proxy law P_X is unchanged by blockwise invertible reparameterizations paired with the pulled-back channels, so the same blockwise non-identification persists under arbitrary conditionally independent measurement noise. $\square$

Proof of Proposition 3 (Narrative collapse as minimax ambiguity). Let $M_1, M_2 \in \mathcal{M}(P_X)$ satisfy $q(M_1) \neq q(M_2)$. Write $a_1 := q(M_1)$ and $a_2 := q(M_2)$ with $a_1 \neq a_2$. Because $M_1, M_2 \in \mathcal{M}(P_X)$, both mechanisms induce the same proxy distribution P_X , and therefore induce the same distribution over datasets $D = (X^{(1)}, \dots, X^{(n)})$, namely P_X^n .

Fix any (measurable) single-valued explanation rule $\hat{a} = \hat{a}(D)$. Since $D \sim P_X^n$ under both M_1 and M_2 , all probabilities below are with respect to the same distribution P_X^n . Define

$$p_1 := \mathbb{P}(\hat{a}(D) = a_1), \quad p_2 := \mathbb{P}(\hat{a}(D) = a_2).$$

Because $a_1 \neq a_2$, the events $\{\hat{a} = a_1\}$ and $\{\hat{a} = a_2\}$ are disjoint, hence $p_1 + p_2 \leq 1$.

Under 0–1 loss $\ell(\hat{a}, q(M)) = \mathbf{1}\{\hat{a} \neq q(M)\}$, the risks at M_1 and M_2 are

$$\mathbb{E}[\ell(\hat{a}(D), q(M_1))] = \mathbb{P}(\hat{a} \neq a_1) = 1 - p_1, \quad \mathbb{E}[\ell(\hat{a}(D), q(M_2))] = 1 - p_2.$$

Therefore,

$$\max \left\{ \mathbb{E}[\ell(\hat{a}(D), q(M_1))], \mathbb{E}[\ell(\hat{a}(D), q(M_2))] \right\} \geq \frac{(1 - p_1) + (1 - p_2)}{2} = 1 - \frac{p_1 + p_2}{2} \geq \frac{1}{2}.$$

Since $\{M_1, M_2\} \subseteq \mathcal{M}(P_X)$, it follows that

$$\sup_{M \in \mathcal{M}(P_X)} \mathbb{E}[\ell(\hat{a}(D), q(M))] \geq \max\left\{\mathbb{E}[\ell(\hat{a}(D), q(M_1))], \mathbb{E}[\ell(\hat{a}(D), q(M_2))]\right\} \geq \frac{1}{2}.$$

Taking the infimum over all single-valued rules $\hat{a}$ yields the stated minimax lower bound:

$$\inf_{\hat{a}} \sup_{M \in \mathcal{M}(P_X)} \mathbb{E}[\ell(\hat{a}(D), q(M))] \geq \frac{1}{2}.$$

In particular, no single-valued rule can be uniformly correct over $\mathcal{M}(P_X)$ whenever two observationally compatible mechanisms disagree on the query. $\square$

Corollary 1 (Scale cannot resolve fixed-regime non-identification) *Let $D_n \sim P_O^n$ be data from a fixed observational regime. If there exist $M_1, M_2 \in \mathcal{M}(P_O)$ such that $q(M_1) \neq q(M_2)$, then for any estimator $\hat{q}_n = \hat{q}_n(D_n)$,*

$$\sup_{M \in \{M_1, M_2\}} \mathbb{P}_M\{\hat{q}_n(D_n) \neq q(M)\} \geq \frac{1}{2}$$

for every n .

Proof. The two observational regimes are identical, so D_n has the same distribution under M_1 and M_2 . Any estimator therefore takes the same distribution of values under both mechanisms. Since $q(M_1) \neq q(M_2)$, the estimator cannot be correct with probability greater than $1/2$ on both simultaneously. Taking the worst case over $\{M_1, M_2\}$ gives the bound. $\square$

C. Off-Support Non-Identification Under Thin Support

Proposition A.1 (Essential non-identification under observational support). Let $\mathcal{X} \subset \mathbb{R}^d$ be a closed embedded k -dimensional submanifold with $k < d$, and assume the observational distribution of X is supported on $\mathcal{X}$. Let $\mathcal{F}$ be a class of scalar-valued functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ closed under smooth, compactly supported perturbations. Then for any $f \in \mathcal{F}$ and any open set U with $U \cap \mathcal{X} = \emptyset$, there exists $g \in \mathcal{F}$ such that $f = g$ on $\mathcal{X}$ but $f \neq g$ on U . Hence f is not identified from observational data on $\mathcal{X}$ without additional assumptions that constrain off-support behavior.

Proof. Because U is open, pick an open ball $B \subset U$. Since $U \cap \mathcal{X} = \emptyset$, then $B \cap \mathcal{X} = \emptyset$. Let ϕ be a smooth bump function supported on B with $\phi(x_0) = 1$ at some $x_0 \in B$. Define $g = f + c\phi$ for any $c \neq 0$. Then $g = f$ on $\mathcal{X}$ and $g(x_0) \neq f(x_0)$. Closure under perturbations implies $g \in \mathcal{F}$. $\square$

The assumption on $\mathcal{F}$ is a convenient sufficient condition; any hypothesis class that can represent (or approximate) localized perturbations exhibits the same qualitative non-identification. This includes rich nonparametric classes and, in practice, overparameterized neural networks that can approximate localized perturbations on off-support regions.

D. Formal Comparison with Rashomon and Uncertainty

Let S denote latent mechanism-level variables, let X denote observed proxies generated by a (possibly stochastic) measurement channel $K(\cdot \mid S)$, and let Y denote an outcome generated by some structural law $Y \sim P(\cdot \mid S)$ or, at the observational level, by the induced conditional $P(Y \mid X)$. The claim concerns *mechanism equivalence under proxies*: for a fixed proxy observational law P_X , define the *mechanism compatibility set* $\mathcal{M}(P_X) = \{M = (P_S, K) : S \sim P_S, X \mid S \sim K(\cdot \mid S) \text{ and the induced marginal is } P_X\}$. When $|\mathcal{M}(P_X)| > 1$, distinct mechanisms $M_1 \neq M_2$ (different latent stories and/or measurement channels) induce the same proxy evidence; thus mechanistic queries $q(M)$ (e.g., counterfactual responses or derivative constraints) are not identified from proxies without additional structure. This is different from the *Rashomon effect*, which is *model-level* multiplicity under a fixed predictive target: for a hypothesis class $\mathcal{H}$ and risk $R(h) = \mathbb{E}[\ell(h(X), Y)]$, the Rashomon set is $\mathcal{R}_\varepsilon = \{h \in \mathcal{H} : R(h) \leq \inf_{h' \in \mathcal{H}} R(h') + \varepsilon\}$, capturing that many predictors achieve essentially the same predictive risk. Rashomon multiplicity lives to the right of X ; the proxy gap lives to the left of X . It is also distinct from *underspecification*, which emphasizes that many predictors can match in-domain objectives yet behave differently under deployment, distribution shift, or interventions: there exist $h_1, h_2 \in \mathcal{R}_\varepsilon$ such that $h_1(X) \neq h_2(X)$ on X drawn from a perturbed or shifted environment. In short, Rashomon and underspecification describe multiplicity *within* a chosen predictor class $\mathcal{H}$ for a fixed observational task (and thus can shrink if one changes $\mathcal{H}$ or adds

regularization), whereas the proxy-gap claim is *mechanism-level non-identification*: even at the population level, and even with infinite data generated through the same measurement channel $K(\cdot | S)$ (so P_X and even $P(Y | X)$ are known exactly), there can remain multiple distinct (P_S, K) (hence multiple incompatible mechanistic answers $q(M)$) consistent with the same proxy evidence. Causal-quartet constructions provide concrete analogues: identical summaries can require different causal adjustment decisions (D’Agostino McGowan et al., 2024), and identical average effects can mask different latent treatment-effect patterns (Gelman et al., 2024).

E. Additional Pea Plant Details and Results

Each MLP is trained with early stopping (patience = 10) and maximum iterations = 250, using random seeds $0, 1, \dots, 59$ (so 60 total fits). The analysis evaluates predictive performance using held-out cross-entropy (log loss) on a 25% test split. Figure 2 visualizes the central identification issue in Regime A: when evidence is reduced to pooled F_2 phenotype proxies, mechanistically distinct explanations can be observationally equivalent on the support of the data. In particular, the Mendelian dihybrid mechanism (dominance with independent segregation of the two traits) and the “independent traits” alternative (fixed marginals $p_R = p_Y = 3/4$) induce essentially the same pooled proxy distribution, so the proxy channel alone does not select a unique mechanism. Figure 3 shows how this equivalence breaks once experimental structure enters: the alternative mechanism is ruled out because it cannot simultaneously explain the outcomes of multiple designed crosses, while the Mendelian mechanism remains well calibrated across `F2_self`, `testcross`, `monoA`, and `monoB`.

Table 2 gives the concrete prediction behind the spread plot. For the `testcross RY×WG`, Mendelian inheritance predicts equal probabilities for the four offspring phenotypes. Without cross-type information, the MLP mean assigns nearly half its mass to `RY`; with `testcross` labels and support, the predictions move close to the Mendelian answer. This is not evidence for a competing biological law, but a symptom of asking a phenotype-only F_2 predictor to answer a counterfactual that its evidence regime does not identify.

Table 2. Concrete counterfactual prediction for `RY×WG` with `cross=testcross`. The Mendelian `testcross` answer assigns probability $1/4$ to each offspring phenotype. The MLP rows report means over near-optimal fits. In the phenotype-only F_2 regime, MLPs are forced to answer an underidentified query and produce an unsupported counterfactual distribution. In the design regime with `testcross` support and labels, predictions move close to the Mendelian target.

	RG	RY	WG	WY
Mendelian <code>testcross</code>	.25	.25	.25	.25
No-design MLP mean	.20	.49	.10	.20
With-design MLP mean	.26	.26	.24	.24

Counterfactual Dispersion and Its Collapse Under Design. Figures 4 and 5 provide an intuition for the spread statistic reported in the main text. Each point cloud displays the predicted offspring phenotype distribution for the same observed parent phenotype pair `RY×WG` across many independently trained MLP fits (restricted to near-optimal seeds). Without cross-type labels and `testcross` support (no design), this counterfactual query is underdetermined: multiple equally predictive models produce noticeably different probability vectors over $\{RG, RY, WG, WY\}$. When the design regime supplies the cross label and `testcross` support, the query becomes well-posed (here, `cross=testcross`), and the counterfactual predictions concentrate, reflecting a smaller mechanism-compatible equivalence class.

Why Restrict to Near-Optimal Seeds. Figure 6 shows the distribution of held-out log loss across MLP random seeds and highlights the near-optimal subset used in Figures 4–5 and in the spread-collapse summary. This addresses the concern that mechanistic disagreement is an optimization artifact: even among predictors with essentially indistinguishable predictive performance, counterfactual mechanistic predictions can diverge when identifying structure is absent, and concentrate when that structure is restored.

F. Simulation Study: In-Domain Agreement, Mechanistic Disagreement

The appendix includes a minimal synthetic study to illustrate that (i) many flexible models can achieve almost identical in-domain prediction on high-dimensional proxies while (ii) disagreeing substantially on mechanistic quantities (derivatives and off-support responses).

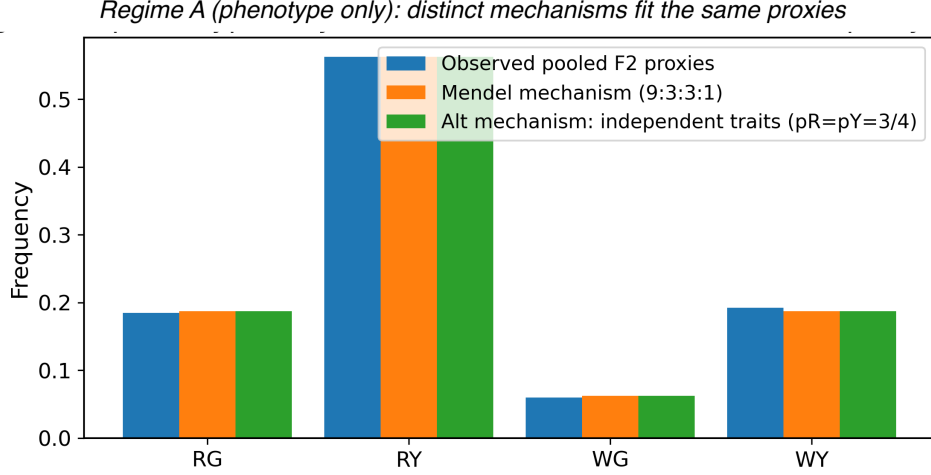


Figure 2. Regime A (phenotype-only): observational equivalence at the proxy level. Pooled F_2 phenotype frequencies over four proxy categories (RG, RY, WG, WY). The Mendelian dihybrid mechanism (dominance + independent assortment; 9:3:3:1) and an alternative “independent traits” mechanism with fixed marginals $p_R = p_Y = 3/4$ are distinct mechanistic stories but induce essentially the same pooled proxy distribution in this regime.

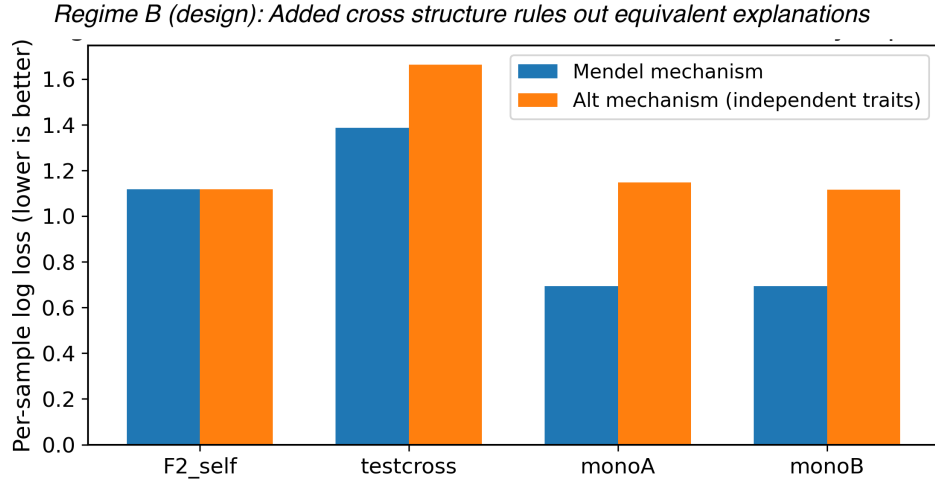


Figure 3. Regime B (design): cross structure rules out an observationally equivalent alternative. Per-sample log loss under additional labeled crosses (F2_self, testcross, monoA, monoB). The alternative mechanism matches Regime A by construction but fails when asked to simultaneously explain the outcomes of multiple designed crosses; the Mendelian mechanism remains well-calibrated across all crosses.

Data-Generating Process. The data-generating process draws $x_1 \sim \mathcal{N}(0, 1)$; sets $x_2 = x_1 + \epsilon_{x,2}$ with $\epsilon_{x,2} \sim \mathcal{N}(0, \sigma_x^2)$; creates correlated proxy blocks $x_3, \dots, x_{d-d_{\text{noise}}}$ via $x_j = \sin(x_1) + 0.3 \cos(0.5x_1) + \epsilon_{x,j}$ with $\epsilon_{x,j} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_x^2)$; and adds d_{noise} pure-noise proxies $\mathcal{N}(0, 1)$. The simulation uses $d = 20$, $d_{\text{noise}} = 6$, $\sigma_x = 0.15$, and $y = \sin(x_1) + 0.1x_2 + \epsilon_y$ with $\epsilon_y \sim \mathcal{N}(0, 0.05^2)$. Train/test sizes are $n_{\text{train}} = 2000$, $n_{\text{test}} = 1000$.

Models. The simulation trains three model families across independent random seeds: (i) MLP (two hidden layers of width 64, ReLU, Adam, 80 epochs, $\text{LR } 2 \times 10^{-3}$), (ii) Random Forest regressor (300 trees, min leaf 5), and (iii) Gradient Boosting regressor (400 trees, depth 3, LR 0.05). Counts: 10 MLPs, 8 RFs, 4 GBRs. All runs use the same data; variation arises only from random seeds (initialization, sampling order, and tree randomness).

Regime A (phentotype only): Counterfactual preds. across near optimal seeds

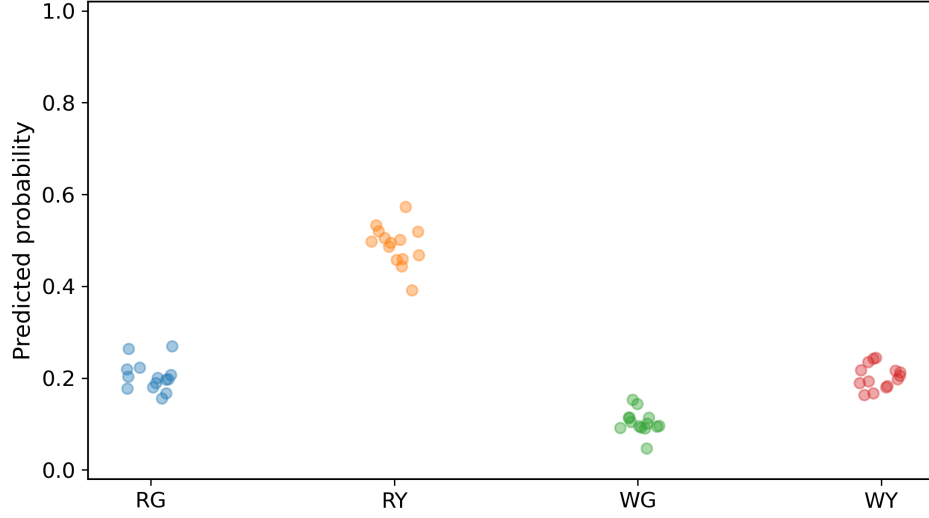


Figure 4. No design label: counterfactual predictions vary across near-optimal black-box fits. Each point is the predicted probability of a phenotype category under a counterfactual query ($RY \times WG$) from a different near-optimal MLP fit trained without cross-type labels. Even among near-optimal predictors, counterfactual mechanistic predictions exhibit nontrivial spread when identifying structure is absent.

Regime B (design): Counterfactual predictions across near optimal seeds

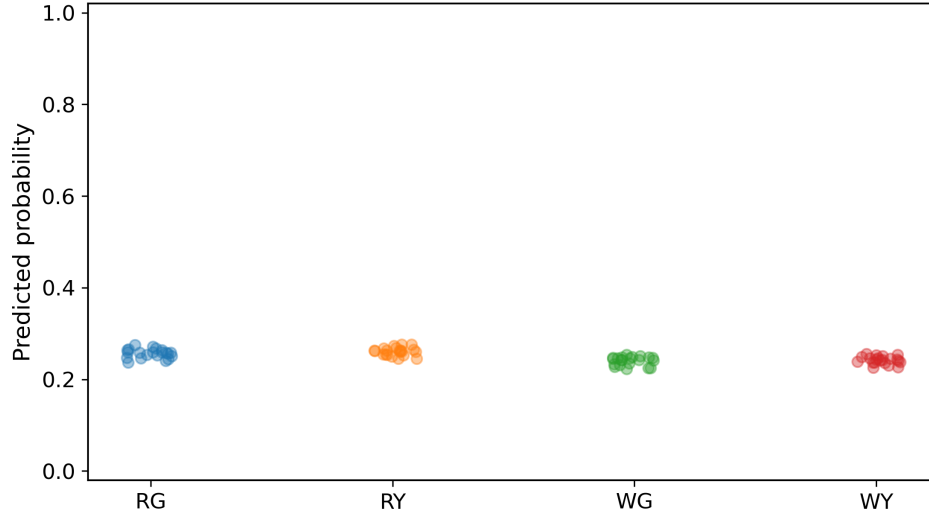


Figure 5. With design label: counterfactual predictions concentrate. Same counterfactual parent phenotype pair as Figure 4, but now the query is well-posed by conditioning on the experimental design label (`cross=testcross`) and models are trained with cross labels. Counterfactual predictions concentrate substantially across near-optimal seeds, reflecting reduced mechanistic ambiguity.

Mechanistic Probes. For each trained model the analysis computes (a) test MSE on an i.i.d. hold-out set; (b) partial derivatives $\partial \hat{y} / \partial x_j$ at 80 test points (autograd for MLPs, finite differences for tree models), summarized by $\text{Var}(\partial \hat{y} / \partial x_j)$ across models; and (c) an off-manifold intervention slice where x_2 is varied in $[-3, 3]$ while other coordinates are held fixed. True gradients are $\partial y / \partial x_1 = \cos(x_1)$, $\partial y / \partial x_2 = 0.1$, and 0 for $j \geq 3$, so variance of order 10^{-3} or below is practically small, while variance $\gtrsim 10^{-2}$ implies meaningfully different mechanistic stories. In the saved simulation output, the displayed gradient-variance values range from roughly 2.4×10^{-5} to 1.2×10^2 , with most values near 10^{-3} and a heavy upper tail driven by tree-model finite-difference gradients on mechanism-bearing and proxy coordinates.

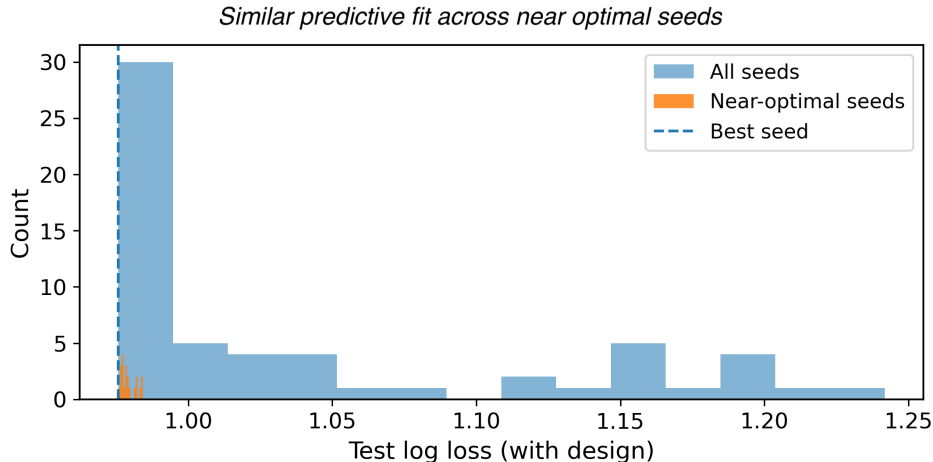


Figure 6. **Black-box multiplicity: many near-optimal fits.** Distribution of held-out test log loss across MLP random seeds when trained with design labels; near-optimal seeds are highlighted. This justifies evaluating mechanistic disagreement *within* a set of comparably good predictors, rather than attributing disagreement to poor optimization.

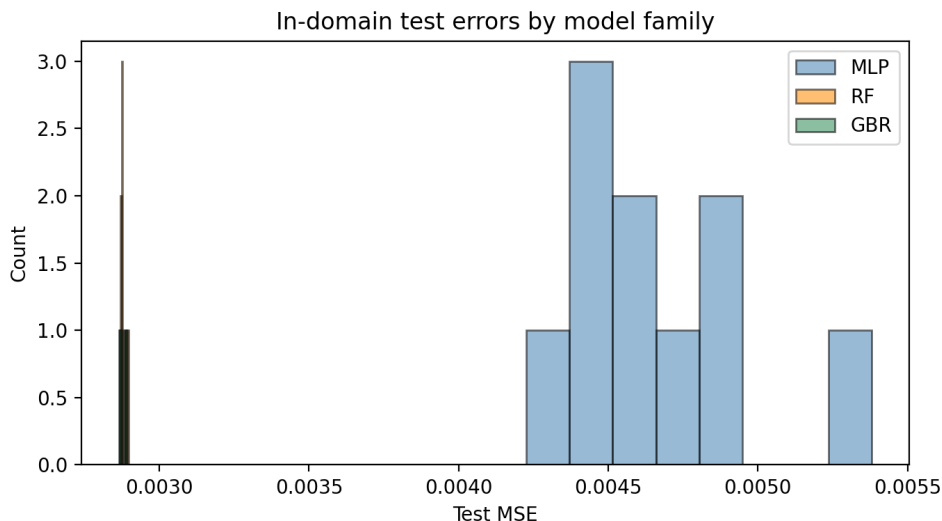


Figure 7. **In-domain fit.** Histogram of test MSE over all seeds and model families. Despite different inductive biases, all models cluster tightly ($\text{MSE} \approx 0.003\text{--}0.005$), showing strong in-domain agreement.

Results. The key pattern is consistent with this paper’s claims: tight predictive clustering (Figure 7) coexists with substantial gradient variance (Figure 8) and divergent off-manifold responses (Figure 9). Because the only differences across runs are random seeds, these disagreements are attributable to underdetermination given proxies and model flexibility, not to differences in data or hyperparameters. The gradient variances provide an interpretable scale: most true gradients are $\mathcal{O}(10^{-1})$, so observed variances $> 10^{-2}$ imply materially different sensitivities and therefore different mechanistic stories, even when predictions on the observed support agree.

G. Six Worked Examples of Identifying Structure

This appendix instantiates the definition in Appendix H in six settings that are representative of modern “Mechanistic AI.” Each example makes the same decomposition explicit:

$$\text{mechanism-side restrictions } (\mathcal{A}_{\text{mech}}) \quad \text{observation-side restrictions } (\mathcal{A}_{\text{obs}}) \quad \text{design / regimes } (r),$$

and points to a mechanistic query $q(M)$ that becomes identifiable only after the relevant structure is imposed.

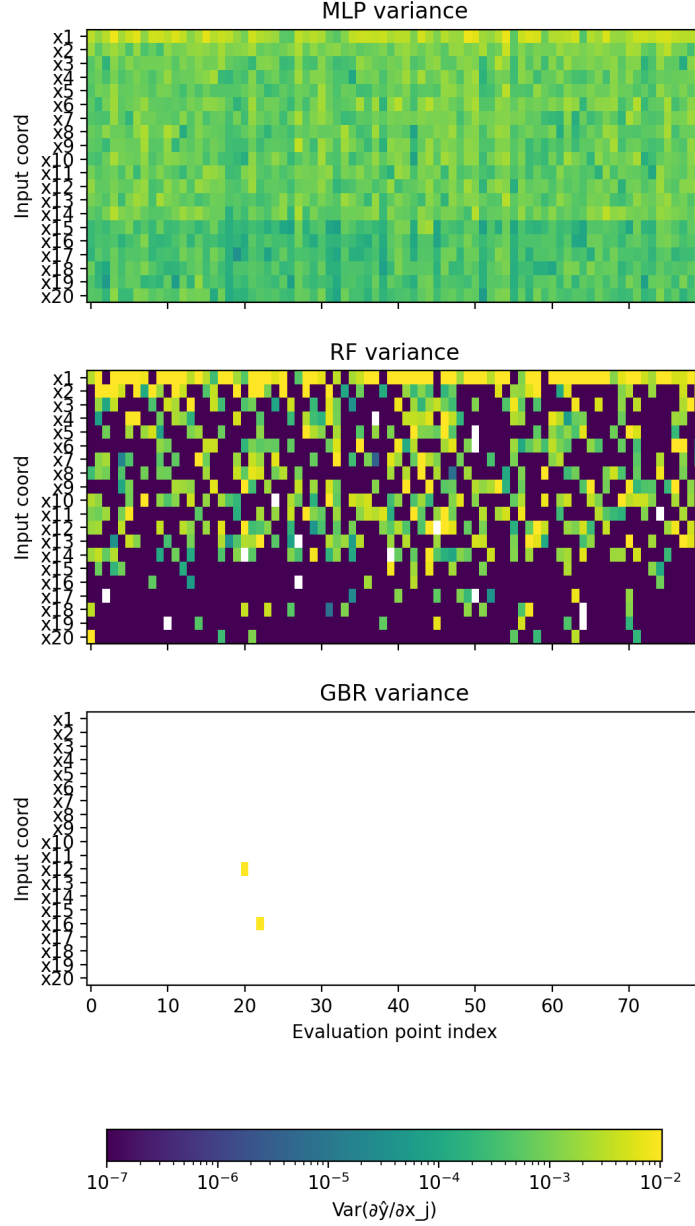


Figure 8. Gradient disagreement by family. Variance of partial derivatives $\partial\hat{y}/\partial x_j$ across seeds within each model family, log-scaled. Dark indicates agreement; bright indicates disagreement. The colorbar (bottom) is the scale for all panels. RF and GBR show sizable variance on x_1 and x_2 (mechanism-bearing coordinates) and spurious variance on proxy/noise coordinates; MLPs exhibit lower but nonzero variance.

Example 1: Disentanglement versus identifiable latent variable models. Locatello et al. show that unsupervised disentanglement is generically non-identifiable without additional inductive bias or supervision; Khemakhem et al. show that identifiability can be restored by adding an observed auxiliary variable and restricting how it modulates the latent distribution (Locatello et al., 2019; Khemakhem et al., 2020).

Latent mechanism $S : S = (S_1, \dots, S_d)$ (“factors”),

Proxies $X : X = f(S)$ (unknown nonlinear mixing),

$\mathcal{A}_{\text{mech}}$: (baseline) $S_1, \dots, S_d$ independent (factorized prior),

$\mathcal{A}_{\text{obs}}$: (baseline) f invertible/smooth (a broad class),

r : (baseline) single environment; observe only X .

$q(M)$: “recover the latent factors” (up to permutation/componentwise transforms).

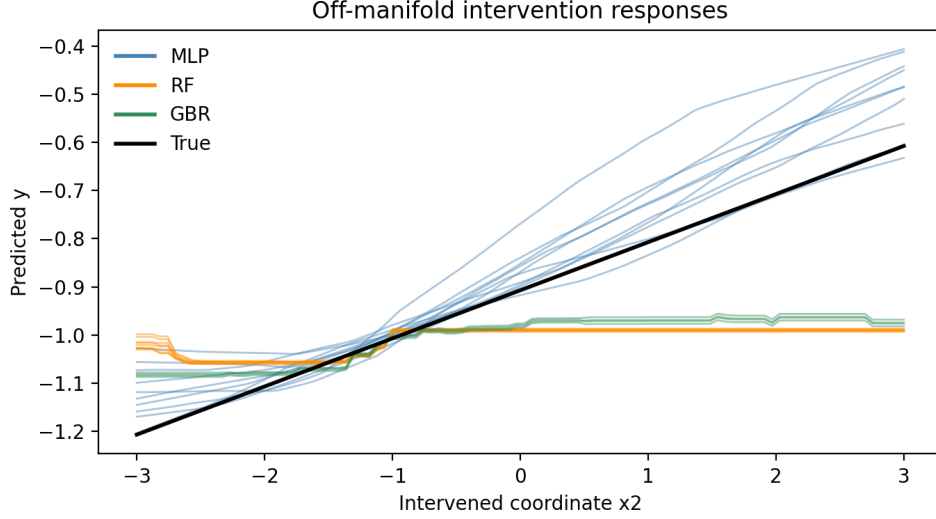


Figure 9. Off-manifold interventions. Each gray line is one model’s prediction when intervening on x_2 (holding other coordinates fixed); black is the true response. Although in-domain MSEs are nearly identical (Figure 7), models diverge markedly off the observed manifold, underscoring underdetermination of mechanism.

Under the baseline regime and assumptions, the equivalence class is large: many distinct latent representations induce the same P_X (non-identification) (Locatello et al., 2019). Identification is recovered by strengthening both observation and design:

$$\begin{aligned} \mathcal{A}'_{\text{mech}} : P_{S|U}(s | u) &= \prod_{j=1}^d \exp\{\eta_j(u)^\top T_j(s_j) - A_j(\eta_j(u))\} \text{ (factorized exponential family),} \\ \mathcal{A}'_{\text{obs}} : X &= f(S) \text{ with suitable regularity/invertibility,} \\ r' : \text{observe an auxiliary variable } U &\text{ (labels/environments/context) that varies } \eta(u). \end{aligned}$$

Then latent variables become identifiable up to simple transformations (the precise equivalence depends on assumptions), turning an ill-posed representation-learning task into an identification problem with explicit structure (Khemakhem et al., 2020).

Example 2: Causal effects with proxy variables for an unmeasured confounder. Miao, Geng, and Tchetgen Tchetgen show that causal effects can be nonparametrically identified in the presence of an unmeasured confounder U if one observes multiple proxy variables with the right conditional-independence and completeness structure (Miao et al., 2018).

Latent mechanism $S : S = (U, \text{(other causes)})$,
Observed (X, Y) : treatment X , outcome Y ,
Proxies (W, Z) : W, Z proxy variables influenced by U ,
 $\mathcal{A}_{\text{mech}}$: structural confounding: $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$,
 $\mathcal{A}_{\text{obs}}$: proxy structure yielding testable conditional independences
 and an operator invertibility/completeness condition,
 r : observe (Y, X, W, Z) in one environment (no intervention required).
 $q(M)$: $P(Y | do(X = x))$ or an average treatment effect.

Here the identifying structure lives crucially in $\mathcal{A}_{\text{obs}}$: observing *two* appropriately related proxies and assuming a completeness/invertibility property gives enough information to identify $q(M)$ even though U is never observed (Miao et al., 2018). The example directly illustrates the paper’s theme: mechanism discovery (a causal effect) is not guaranteed by flexible prediction, but can become possible when the *measurement channel* is structured and stated explicitly.

Example 3: Causal representation learning from grouped observables. Morioka and Hyvärinen establish identifiability for causal representation learning by imposing a grouping/modularity structure on the observation process: observational variables can be partitioned

into groups that each depend on a particular latent factor (Morioka & Hyvarinen, 2024).

Latent mechanism $S : (S_1, \dots, S_M)$ with a causal model among blocks,
Proxies $X : X = (X_{G_1}, \dots, X_{G_M})$ partitioned into observed groups G_m ,
 $\mathcal{A}_{\text{mech}}$: a causal model among $S_1, \dots, S_M$ (allowing confounders/cycles under stated conditions),
 $\mathcal{A}_{\text{obs}}$: X_{G_m} depends (primarily) on S_m
 via a group-specific mixing/channel; groups are modular,
 r : single environment; observe all groups (the key is the grouped measurement structure).
 $q(M)$: latent causal structure and/or latent features (up to allowable reparameterizations).

This is a clean example where identification is purchased mainly by *measurement modularity*: the grouping assumption shrinks $\mathcal{M}(P_X)$ by ruling out alternative mixings that would otherwise be observationally equivalent (Morioka & Hyvarinen, 2024). For a precise theorem in this direction, see Theorem 1 of Morioka & Hyvarinen (2024). Appendix I provides a complementary, more general template for block identification and orientation under grouped observables plus additional asymmetries.

Example 4: Orientation and causal features from invariance across environments (ICP). Invariant causal prediction treats identification as an invariance-matching problem across environments: causal parents are those predictors for which the conditional law of Y given those predictors remains stable across environments (Peters et al., 2016).

Latent mechanism S : a causal model generating (X, Y) ,
Observed $(X^{(e)}, Y^{(e)})$: data collected across environments $e \in \mathcal{E}$,
 $\mathcal{A}_{\text{mech}}$: $\exists S^* \subseteq \{1, \dots, p\}$ s.t. $P(Y^{(e)} \mid X_{S^*}^{(e)})$ is invariant in e ,
 $\mathcal{A}_{\text{obs}}$: environment labels are observed;
 shifts affect non-causal parts or inputs but preserve the structural law,
 r : multi-environment data collection (natural experiments, domain shifts, designed perturbations).
 $q(M)$: S^* (a set of causal predictors/parents, or constraints implied by invariance).

This example emphasizes the role of design: identification improves because r supplies variation that breaks observational equivalence. More i.i.d. data from a single environment would not create the invariance contrasts that ICP exploits. This example presents a template, not a new theorem: it synthesizes the ICP invariance principle (Peters et al., 2016) with grouped-observable latent-structure results such as Morioka & Hyvarinen (2024).

Example 5: Graph-based identification of interventions (do-calculus and the ID algorithm). In causal inference, a causal graph plus intervention semantics can turn an observational distribution into an identifiable causal quantity. Pearl’s framework formalizes interventions via $do(\cdot)$, and Shpitser and Pearl provide a complete identification procedure (the ID algorithm) for many settings with hidden variables (Pearl, 2009; Shpitser & Pearl, 2006).

Mechanism M : a structural causal model over variables V (possibly with latent confounders),
Observed $P(V)$: the observational distribution over measured nodes,
 $\mathcal{A}_{\text{mech}}$: causal Markov property w.r.t. a given graph G
 and intervention semantics (edge cutting),
 $\mathcal{A}_{\text{obs}}$: the measured variables correspond to nodes in G ;
 unmeasured causes are represented as latent nodes,
 r : may be purely observational given G , or strengthened by collecting additional nodes / interventions.
 $q(M)$: $P(Y \mid do(X = x))$ or more general interventional distributions.

Here, identifying structure is explicit and audit-ready: it is exactly the graph and intervention semantics plus any assumptions about which variables are measured. This is the econometric identification mindset in its most literal form: what is learned is not “the mechanism” in the abstract, but a particular interventional query that is identifiable under stated structure.

Example 6: Equation discovery in physical systems (symbolic regression, sparse dynamics, PINNs). A large class of “scientific ML” methods succeed because they impose strong structure on both the mechanism and the measurement process (Schmidt & Lipson, 2009;

Brunton et al., 2016; Raissi et al., 2019). The details differ, but the identifying logic is the same.

Mechanism M : a low-complexity governing law (ODE/PDE) with constrained form,

Observed X : measurements of system states (and sometimes derivatives) under selected conditions,

$\mathcal{A}_{\text{mech}}$: a restricted hypothesis class: sparsity, conservation,

known operator library, smoothness,

$\mathcal{A}_{\text{obs}}$: measurement access to the relevant state variables and/or their derivatives,

with controlled noise/coarsening,

r : designed initial/boundary conditions, forcing, or experimental settings that excite informative variation.

$q(M)$: the governing equation/operator and/or its parameters (and implied invariances/derivatives).

Concretely, symbolic regression restricts the form of candidate laws and searches for simple expressions consistent with data (Schmidt & Lipson, 2009). Sparse identification (SINDy) assumes that $\dot{x}(t) = \Theta(x(t))\xi$ for a known feature/library map $\Theta(\cdot)$ with a sparse coefficient vector ξ , and identification depends on measuring (or stably approximating) derivatives and on trajectories that sufficiently “excite” the library (Brunton et al., 2016). Physics-informed neural networks incorporate the constraint that a PDE residual must vanish and leverage boundary/initial conditions as additional structure; inverse problems remain identification problems whose solvability depends on what is observed and what regimes are included (Raissi et al., 2019). Across these methods, the moral is identical to the paper’s: mechanistic success comes from explicit, checkable identifying structure—not from black-box flexibility alone.

H. A Formal Definition of Identifying Structure

This appendix makes precise the meaning of “identifying structure” and why it necessarily has *three* components: assumptions about the latent mechanism, assumptions about the proxy process, and assumptions about design/data collection. The formal definition presents this as two assumption bundles (one for the mechanism and one for the process), combined with a design/data collection regime. The definition is query-specific: identification is always relative to a mechanistic question q (e.g., a counterfactual effect, a derivative sign, or an invariance claim).

Mechanisms and observational regimes. Let $\mathcal{M}$ be a set of candidate mechanisms. An element $M \in \mathcal{M}$ should be understood broadly as a complete mechanistic story, including: (i) latent mechanism-level variables S with some law P_S , (ii) an observation or measurement channel K generating proxies X from S , and (iii) any additional structural laws needed to answer mechanistic queries (e.g., an outcome law $P(Y | S)$, or a causal model supporting interventions). Write this schematically as

$$M = (P_S, K, (\text{structural laws} / \text{intervention semantics})).$$

Data collection enters through an *observational regime* r . The regime specifies which environments or interventions are observed and what variables are measured in each. Formally, let $\mathcal{E}$ index environments (which may include explicit interventions), and let $O^{(e)}$ denote the observed variables in environment e (often $O^{(e)} = (X^{(e)}, Y^{(e)})$). The regime induces an *observation operator*

$$T_r : \mathcal{M} \rightarrow \prod_{e \in \mathcal{E}} \mathcal{P}(\mathcal{O}^{(e)}), \quad T_r(M) = \{P_O^{(e)}(M)\}_{e \in \mathcal{E}},$$

mapping each mechanism M to the family of observable distributions it implies under the regime. The single-environment proxy-only setting in the main text corresponds to $\mathcal{E} = \{e_0\}$ and $O^{(e_0)} = X$, so $T_r(M) = P_X(M)$.

Assumptions split into mechanism-side and observation-side. Let $\mathcal{A}$ denote a set of restrictions on mechanisms, i.e., a subset $\mathcal{A} \subseteq \mathcal{M}$. To reflect the logic in the paper, decompose restrictions into

$$\mathcal{A} = \mathcal{A}_{\text{mech}} \cap \mathcal{A}_{\text{obs}},$$

where $\mathcal{A}_{\text{mech}}$ constrains the latent mechanism and its interventional semantics (e.g., causal graph, functional form, invariances, monotonicities, sparsity, conservation laws), and $\mathcal{A}_{\text{obs}}$ constrains the measurement process linking S to X (e.g., grouping/modularity, conditional independence, completeness/informativeness, known measurement error structure, anchor variables, repeated measurements).

Compatibility sets and identified sets. Fix an observational regime r and an observed family of distributions $\{P_O^{(e)}\}_{e \in \mathcal{E}}$ (the population object that the dataset estimates). The set of mechanisms compatible with this evidence under assumptions

$\mathcal{A}$ is

$$\mathcal{M}(\{P_O^{(e)}\}_{e \in \mathcal{E}}; \mathcal{A}, r) = \left\{ M \in \mathcal{A} : T_r(M) = \{P_O^{(e)}\}_{e \in \mathcal{E}} \right\}.$$

This generalizes the main-text compatibility set $\mathcal{M}(P_X)$, which corresponds to $\mathcal{A} = \mathcal{M}$ and a regime with one environment and $O = X$.

Let $q : \mathcal{M} \rightarrow \mathcal{Q}$ be a mechanistic query (e.g., a causal effect, an intervention response curve, a derivative sign, an invariance statement). The *identified set* of answers under $(\mathcal{A}, r)$ is

$$\mathcal{I}_q(\{P_O^{(e)}\}_{e \in \mathcal{E}}; \mathcal{A}, r) := \left\{ q(M) : M \in \mathcal{M}(\{P_O^{(e)}\}; \mathcal{A}, r) \right\}.$$

The query q is *point-identified* under $(\mathcal{A}, r)$ if $\mathcal{I}_q(\cdot)$ is a singleton; otherwise q is only *partially identified*, and the width/geometry of $\mathcal{I}_q(\cdot)$ is the population-level object corresponding to the *structural identification uncertainty* emphasized in the paper.

Definition (Identifying structure). Fix a query q and an observational regime r . An *identifying structure* for q (at the population level) is any pair

$$(\mathcal{A}_{\text{mech}}, \mathcal{A}_{\text{obs}}) \text{ together with } r$$

such that the identified set $\mathcal{I}_q(\{P_O^{(e)}\}; \mathcal{A}, r)$ is a singleton (or, in applications, is sufficiently small to support the intended scientific claim). The key point is that identifying structure is inherently joint: mechanism-side restrictions alone do not identify q if measurement can be reparameterized to preserve the same proxies, and observation-side restrictions alone do not identify q if the mechanism class remains too broad. Discriminating data collection is represented by r : changing environments, adding interventions, or adding measurement channels can change the family $\{P_O^{(e)}\}$ and shrink the compatibility set, even when “more data” within a fixed environment cannot.

Appendix I instantiates this definition for a particular target: identifying a modular *block* structure of latent variables (up to within-block reparameterizations) and orienting relations among blocks. The point of the definition above is broader: it cleanly separates (i) what is assumed about mechanisms, (ii) what is assumed about measurement, and (iii) what is purchased by design.

I. Minimal Structure for Identifiability (Blocks and Orientation)

Sections 3–5 emphasize non-identification: from proxies alone, many mechanistically distinct stories can be observationally indistinguishable. This appendix records a complementary point that is essential for a non-nihilist position: *identifiability becomes possible* once one states and defends additional structure. The goal here is not to develop a full theory, but to give a minimal, checkable template for when (i) a modular *block* structure of latent drivers can be recovered, and (ii) *directions* among blocks can be oriented. The results in this appendix are stated as templates: they summarize the kinds of additional assumptions under which identifiability can be obtained, rather than presenting new identifiability theorems proved in this paper.

Setup. Let $S = (S_1, \dots, S_B)$ denote latent mechanism variables (“blocks”), and let $X = (X_1, \dots, X_p)$ denote observed proxies. Assume proxies are generated from S by either a deterministic measurement map $X = f(S)$ or a stochastic measurement channel $X | S \sim K(\cdot | S)$. As in the main text, identification is understood *up to within-block reparameterizations*: for invertible g_b , replacing S_b by $g_b(S_b)$ and pulling back the measurement channel leaves P_X unchanged.

A. Block structure from grouped observables

The strongest form of “minimal structure” is a *grouping* assumption: each subset of observed variables is driven primarily by one latent block, and cross-block relations show up as dependence *between* groups.

Assumption A1 (Grouping / modular measurement). There exists a partition of proxy coordinates into B nonempty groups, $\{1, \dots, p\} = \bigsqcup_{b=1}^B G_b$, such that for each b the group X_{G_b} depends on the latent block S_b through a group-specific map or channel:

$$X_{G_b} | S \sim K_b(\cdot | S_b), \quad \text{and} \quad K(\cdot | S) = \prod_{b=1}^B K_b(\cdot | S_b).$$

(Deterministic measurement is included by taking K_b to be a point mass at $f_b(S_b)$.)

Assumption A2 (Channel informativeness). Each group-channel K_b is informative enough that distinct functions of S_b cannot be “hidden” completely by measurement. A standard sufficient condition is an injectivity/completeness-type property: if $\varphi(S_b)$ is nontrivial, then its footprint is detectable from X_{G_b} (e.g., the conditional expectation operator $\varphi \mapsto \mathbb{E}[\varphi(S_b) \mid X_{G_b} = \cdot]$ is injective on a rich class).

Assumption A3 (Non-separable cross-block dependence). The joint law of S is not additively separable across the true blocks. One convenient way to state this is graphically: there exists an undirected graph H on $\{1, \dots, B\}$ such that for each edge $\{b, b'\}$, S_b and $S_{b'}$ exhibit genuine cross-block dependence (not removable by within-block reparameterizations), and H is connected.

Template A.1 (Block identifiability up to within-block reparameterization). Under Assumptions A1–A3, the grouping of proxies into latent blocks is identifiable from P_X up to permutation of the block labels and within-block reparameterizations. In particular, any alternative representation that preserves P_X must correspond to the same coarsest partition of proxy coordinates into conditionally independent groups given the latent blocks, modulo relabeling.

Discussion. This template is intentionally stated at a high level: it isolates what must be made explicit to move from “many mechanisms fit the proxies” to “the proxies pick out a modular decomposition.” The critical ingredients are (i) a defensible measurement grouping, (ii) informative channels so that group-level signals are not annihilated by coarsening/noise, and (iii) cross-block dependence strong enough that blocks cannot be merged or split without changing P_X . Recent causal representation learning results formalize identifiability from such grouped-observable structure and show consistent estimation under concrete generative conditions.

B. Orientation: two minimal routes

Once blocks are identified up to reparameterization, orienting relations among blocks requires *additional asymmetry*. Two common sources are (i) *multiple environments* (invariance under interventions/shifts) and (ii) *signed asymmetries* in the joint law.

Route 1: Orientation from invariances across environments. Suppose the data come from environments $e \in \mathcal{E}$ with proxy distributions $P_X^{(e)}$ induced by latent laws $P_S^{(e)}$ and the *same* measurement channels K_m . Assume the causal mechanism among blocks is invariant while some environments perturb the distribution of certain blocks (e.g., shift interventions).

Assumption B1 (Invariant mechanism, shifting inputs). There exists a directed acyclic graph (DAG) G over blocks $S_1, \dots, S_B$ such that for each environment e the conditional law of each block given its parents in G is invariant in e , while the marginal law of some blocks varies with e .

Template A.2 (Orientation from invariance). Under Assumptions A1–A3 and B1 (plus mild regularity so that invariances are testable at the level of X via the informative channels), the parent sets in G are identifiable (up to the same within-block reparameterizations), hence the edges of G can be oriented by selecting the unique DAG whose implied invariance statements match the observed family $\{P_X^{(e)}\}_{e \in \mathcal{E}}$.

Discussion. This is the canonical “scientific” route: orientation is purchased by *variation across environments* that leaves the mechanism invariant. In practice it corresponds to: collect (or exploit) multi-environment data, then test which conditional relations remain stable across environments.

Route 2: Orientation from a sign-faithful observable asymmetry. Alternatively, direction can be identified when the observed law contains an asymmetry whose sign is preserved by the admissible measurement transformations and tracks the arrow direction on an already identified skeleton.

Assumption B2’ (Observable sign-faithful orientation score). For each adjacent unordered pair of identified blocks $\{b, b'\}$ in the skeleton H , there exists an observable measurable functional

$$\Delta_{bb'} = \Delta_{bb'}(X_{G_b}, X_{G_{b'}})$$

such that: (i) $\Delta_{bb'} = -\Delta_{b'b}$ a.s.; (ii) $\Delta_{bb'}$ is invariant under admissible within-block reparameterizations and preserved by

the measurement model; and (iii) whenever b and b' are adjacent,

$$\mathbb{E}[\Delta_{bb'}] > 0 \iff b \rightarrow b', \quad \mathbb{E}[\Delta_{bb'}] < 0 \iff b' \rightarrow b,$$

and in particular $\mathbb{E}[\Delta_{bb'}] \neq 0$.

Proposition 4 (Orientation from a sign-faithful observable asymmetry) *Assume A1–A3, and suppose the undirected skeleton H over blocks has been identified. Under Assumption B2', each adjacent edge $\{b, b'\}$ in H is oriented by the sign of $\mathbb{E}[\Delta_{bb'}]$: it is oriented as $b \rightarrow b'$ if $\mathbb{E}[\Delta_{bb'}] > 0$, and as $b' \rightarrow b$ if $\mathbb{E}[\Delta_{bb'}] < 0$.*

Proof. Fix an adjacent edge $\{b, b'\}$ in the identified skeleton H . By Assumption B2', the functional $\Delta_{bb'}$ is observable and invariant under admissible within-block reparameterizations, so the sign of $\mathbb{E}[\Delta_{bb'}]$ is determined by the observed law of $(X_{G_b}, X_{G_{b'}})$ and does not depend on the choice of latent coordinates within each block. The sign-faithfulness clause implies

$$\mathbb{E}[\Delta_{bb'}] > 0 \iff b \rightarrow b', \quad \mathbb{E}[\Delta_{bb'}] < 0 \iff b' \rightarrow b.$$

Because $\mathbb{E}[\Delta_{bb'}] \neq 0$, exactly one of these cases holds, so the sign of $\mathbb{E}[\Delta_{bb'}]$ orients the edge uniquely. The argument is edgewise, so the result is local to adjacent edges in H . $\square$

These templates are not claims that identifiability should always be expected; rather, they illustrate that claims of mechanism discovery must be paired with an explicit statement of *which* identifying structure is being assumed (grouping, invariances across environments, anchors/asymmetries, etc.) and *how* it is being tested. Absent such structure, Sections 3–5 apply: proxy-rich, high-dimensional settings generically leave large mechanism equivalence classes, and fluent single-narrative explanations are epistemically unsafe.